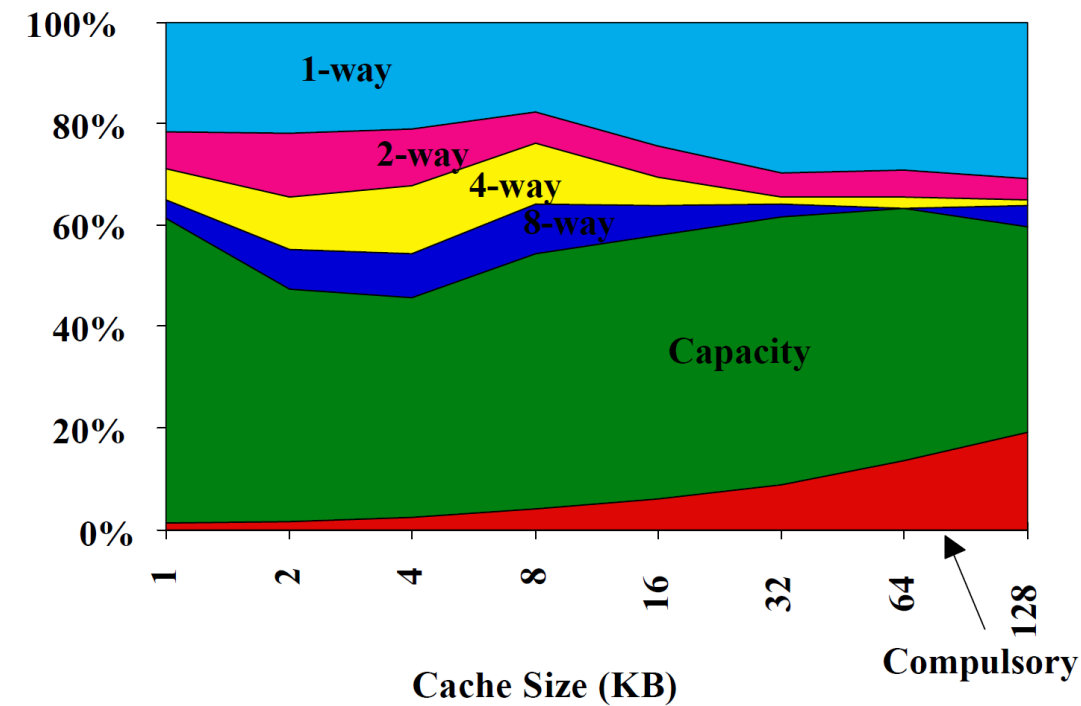


CACHE MISS AND WAYS TO REDUCE CACHE MISS RATE

• Understanding the types of cache miss: The Three C's + 'C'

- **Compulsory (Cold-start)**: the first access to a block in cache
- **Capacity**: desired data cannot be contained in the cache (size limitation)
- **Conflict (Collision)**: desired data conflicts with other content in the cache.
- **Coherence**: miss due to external invalidations (multiprocessor architecture)



Question: How do we minimize these Cache Misses?

Rather, How do we better utilize these caches?

Reduce the Miss Rate and/or **Reduce the Miss Penalty**
May Also Reduce the Hit Time

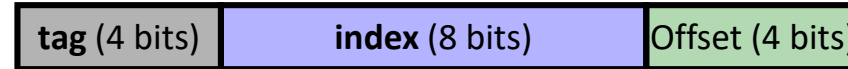
16-BIT ADDRESS, 4KB DM CACHE, 16B BLOCKS

Main mem (64KB) Arrays A, B, C are placed contiguously in memory.

Q0. Start address of a[] is **0x0000**, b[]=0x____;c[]= 0x____.

```
#define N 256
```

```
uint8_t a[N], b[N], c[N];
for (register i = 0; i < N; i++)
    c[i] = a[i] + b[i];
```



Q1. Number of Blocks/Sets = ____ and data bytes per block ____.

Q2. Total memory/cache access made to run this code ____.

Q3. Total Cache Hits ____ and Cache Misses ____.

- | | |
|--------------|---------------|
| 1. Load a[0] | 3. Add |
| 2. Load b[0] | 4. Store c[0] |

0x0000	0
0x0001	1
0x0002	2
0x0003	3
0x0004	4
0x0005	5
	:
	:
	:
	:
	:
0xFFFFC	M
0xFFFFD	N
0xFFFFE	P
0xFFFFF	Q

		16 B															
256	0x00	M	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H
	0x01	M	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H
	0x02																
	0x03																
	:																
	0x10	M	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H
	0x11	M	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H
	0x12																
	:																
	0x20	M	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H
	0x21	M	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H
	:																
	0xFF																

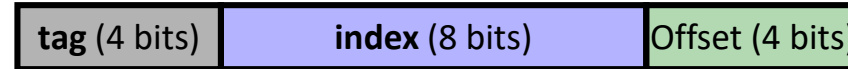
16-BIT ADDRESS, 4KB DM CACHE, 16B BLOCKS

Main mem (64KB) Arrays A, B, C are placed contiguously in memory.

Q0. Start address of a[] is **0x0000**, b[]=0x____;c[]= 0x____.

```
#define N 1024
```

```
unit16_t a[N], b[N], c[N];
for (register i = 0; i < N; i++)
    c[i] = a[i] + b[i];
```



Q1. Number of Blocks/Sets = ____ and data bytes per block ____.

Q2. Total memory/cache access made to run this code ____.

Q3. Total Cache Hits ____ and Cache Misses ____.

- | | |
|--------------|---------------|
| 1. Load a[0] | 3. Add |
| 2. Load b[0] | 4. Store c[0] |

0x0000	0
0x0001	1
0x0002	2
0x0003	3
0x0004	4
0x0005	5
	:
	:
	:
	:
	:
0xFFFC	M
0xFFFD	N
0xFFFE	P
0xFFFF	Q

